DIRECTORATE OF GOVERNMENT EXAMINATIONS, CHENNAI-6

HSE SECOND YEAR EXAMINATIONS - MARCH-2025

BUSINESS MATHEMATICS AND STATISTICS - ANSWER KEY

MAXIMUM MARKS : 90

GENERAL INSTRUCTIONS

1. Answers written only in BLACK or BLUE ink should be evaluated
2. For objective type questions, award 1 mark for "writing the correct option's code and the corresponding option's answer"
3. Marks should be awarded for suitable alternative method also.
4. Mark(s) should not be reduced for the correct answer/stage, if it is written without formula/properties also, 2 * means award one mark for the formula.
5. Award full mark directly, if the solution is arrived with no mistakes without giving weightage for the stages.
6. The stage mark is essential, only if the part of the solution is incorrect.
7. Award marks, if the answer is in decimal value and also approximately equal to the key answer.
8. I**mportant Note for Part II, Part III and Part IV**

For a particular stage in which the stage mark is greater than 1 and one who begins with correct step but reaches with incorrect solution, for such suitable credits should be given by breaking the stage marks.

PART – I

i. Answer all the questions
ii. Choose the Most Appropriate from the given Four Alternatives and write the option code and the corresponding answer

20×1=20

Q. No	Option	Answer	Mark
1.	(b)	$1/2 (\log x)^2 + C$	1
2.	(a)	28 /3	1
3.	(d)	1	1
4.	(a)	$\lvert A \rvert^{n-1}$	1
5.	(b)	MR	1
6.	(a)	2 sq.Units	1
7.	(d)	Of order 1 and degree 3	1
8.	(d)	$\frac{1}{x}$	1
9.	(b)	Both equal and unequal intervals	1

10.	(a)	$1 - E^{-1}$	**1**
11.	(d)	Cumulative Probability	**1**
12.	(a)	$V(x)$	**1**
13.	(c)	$4\left(\frac{2}{3}\right)^6$	**1**
14.	(c)	$\frac{1}{81}$	**1**
15.	(d)	Random sample	**1**
16.	(b)	Two types	**1**
17.	(d)	All the above	**1**
18.	(b)	Fisher's	**1**
19.	(a)	VAM	**1**
20.	(b)	Total Supply $\neq$ Total Demand	**1**

PART – II

Answer **any Seven** questions. Question No.**30** is **compulsory.** **7×2 = 14**

Q.No	Answer	Mark
21	$Order\ of\ A\ \ is\ 3\ \times 4$ $\rho(A\ ) \leq 3$ $A \sim \begin{pmatrix} 1 & 2 & -1 & -5 \\ 0 & 7 & -8 & 14 \\ 0 & 0 & 0 & -7 \end{pmatrix}$ $\rho(A) = 3$ (* Any suitable method can be awarded full marks)	**1** **1**
22.	$\int \frac{dx}{sin^2x\ \cos^2 x}$ $\int \frac{sin^2x + \cos^2 x}{sin^2x\ \cos^2 x}\ dx$ $\int (\sec^2 x + cosec^2\ x)\ dx$ $= \tan x - \cot x + C$	**1** **1**
23.	$MR\ =\ 9 - 4x^2$ $R = 9x - 4\frac{x^3}{3}$ $Demand\ Function = p = \frac{R}{x}$ $p = 9 - 4\frac{x^2}{3}$	**1** **1**
24.	$\sum p_i = 1$ (or) 2k+3k+4k=1 K = $\frac{1}{9}$	**1** **1**

25.

$$x_{ij} = \begin{cases} 1 & if\ i^{th}\ job\ is\ assigned\ to\ j^{th}\ machine \\ 0 & if\ i^{th}\ job\ is\ not\ assigned\ to\ j^{th}\ machine \end{cases}$$

2

26.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	-1				
		1			
1	0		4		
		5		6	
2	5		10		0
		15		6	
3	20		16		0
		31		6	
4	51		22		
		53			
5	104				

For finding y values award 1 Mark + Remaining part 1 Mark

2

27.

np = 20 , $\sqrt{npq} = 4$

q = 4/5

p = 1/5

n = 100

1

1

28.

P = 0.575 , Q = 0.425

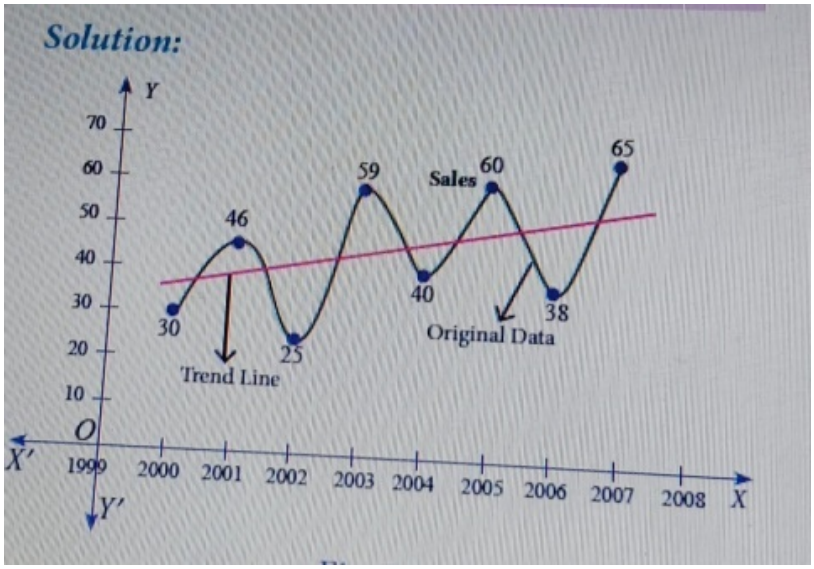
S.E = $\sqrt{\frac{PQ}{n}}$

= $\sqrt{\frac{0.575\ X\ 0.425}{400}} = 0.025$

1

1

29.



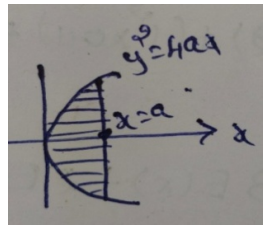
2

30.	Auxillary Equation $m^2 + 2m + 2 = 0$ $\alpha = -1 \quad , \quad \beta = 1$ y = $e^{\alpha x}$ (A $\cos \beta x$ + B $\sin \beta x$) y= e^{-x} (A $\cos x$ + B $\sin x$)	**1** **1**

PART - III

Answer **any Seven** questions. Question No.**40** is **compulsory.** **7×3=21**

Q.No	Answer	Mark
31.	11 x + 3y = 64 8 x + 3y = 49 $\Delta = \begin{vmatrix} 11 & 3 \\ 8 & 3 \end{vmatrix} = 9 \neq 0$ $\Delta x = \begin{vmatrix} 64 & 3 \\ 49 & 3 \end{vmatrix} = 45$ $\Delta y = \begin{vmatrix} 11 & 3 \\ 8 & 3 \end{vmatrix} = 27$ $x = 5, \; y = 3$	**1** **1** **1**
32.	u = log x , v = $\frac{x^2}{2}$ $\int u dv = uv - \int v du$ $\int x log x \, dx = \frac{x^2}{2} \, log x - \frac{x^2}{4} + c$ $\int x log x \, dx = \frac{x^2}{2} (\log x - \frac{1}{2}) + C$	**1** **1** **1**

33.	$y^2 = 4ax$ $y = 2\sqrt{a}\sqrt{x}$  Area = $\int_a^b y\,dx$ Area = **2** $\int_0^a 2\sqrt{a}\sqrt{x}dx$ = $\frac{8}{3}a^2$ sq units	**1** **1** **1**
34.	$y = \frac{a}{x} + b$ $\frac{dy}{dx} = \frac{-a}{x^2}$ $x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$ $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$	**1** **2**
35.	$\Delta^4 y_0 = 0$ $(or)\ (E-1)^4 y_0 = 0$ $(or)\ (E^4 - 4E^3 + 6E^2 - 4E + 1)y_0 = 0$ $(or)\ y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$ $67.4 - 4y_3 + 6(54.1) - 4(49.2) + 45 = 0$ $y_3 = 60.05$	1 1 1
36.	$E(x) = \sum x_i P_i = 1.8$ $E(x^2) = \sum x^2 P_i = 4.4$ $E(3x + 2x^2) = 3E(x) + 2E(x^2)$ $= 3(1.8) + 2(4.4)$ $= 14.2$	**1** **1** **1**
37.	n=7 $p=q = \frac{1}{2}$ $P(X = x) = n^C\ x\ p^x\ q^{n-x}$ $P(X = 2) = 7^C\ 2(\frac{1}{2})^2\ (\frac{1}{2})^{7-2} = \frac{21}{128} = 0.164$	1 1 1

38.

n = 9000

P = 0.3333 , Q = 1-P = 0.6667 **1**

$SE = \sqrt{\frac{PQ}{n}}$ **1**

$= \sqrt{\frac{0.3333 \; x \; 0.6667}{9000}}$ = 0.00496 **1**

39.

Type of Shampoo	Estimated Sales in Rupee			Minimum	Maximum
	15000	**10000**	**5000**		
Egg Shampoo	30	10	10	10	30
Clinic Shampoo	40	15	5	5	40
Deluxe Shampoo	55	20	3	3	55

1

Max of (10, 5, 3) = 10 **1**

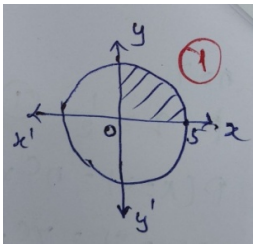
Marketing Manager's Decision is sell Egg Shampoo

Mini of (30, 40, 55) = 30 **1**

Egg Shampoo must be launched.

40.

$x^2 + y^2 = 25$



1

Area = $\int_a^b y dx$ **1**

Area = $4 \int_0^5 y dx$

$= 4 \int_0^5 \sqrt{25 - x^2}$ dx

= 25 π sq unit **1**

PART - IV

Answer All the Questions :- **7×5=35**

41. a	$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 \\ 52 \\ 0 \end{pmatrix}$	1
	$AX = B$	
	$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 2 & 5 & 7 & : & 52 \\ 2 & 1 & -1 & : & 0 \end{bmatrix}$	1
	$A:B \sim \begin{pmatrix} 1 & 1 & 1 & 9 \\ 0 & 3 & 5 & 34 \\ 0 & 0 & -4 & -20 \end{pmatrix}$	
	$\rho(A) = \rho(A,B) = 3$	1
	$\left.\begin{aligned} x+y+z &= 9 \\ 3y+5z &= 34 \\ -4z &= -20 \end{aligned}\right\}$	1
	$x = 1, \quad y = 3, \quad z = 5$	1
	(OR)	
41. b	$I = \dfrac{3x^2-2x+5}{(x-1)(x^2+5)} = \dfrac{A}{x-1} + \dfrac{Bx+c}{x^2+5}$	1
	$A = 1, B = 2, C = 0$	1
	$= \int \left(\dfrac{1}{x-1} + \dfrac{2x}{x^2+5}\right) dx$	1
	$\left.\begin{aligned} &= \log\lvert x-1\rvert + \log\lvert x^2+5\rvert + c \\ &\text{(or) } \log[(x-1)(x^2+5)] + c \\ &\text{(or) } \log(x^3-x^2+5x-5) + c \end{aligned}\right\}$	2

42. a

$P = 2\tan x \ , \quad Q = sin\ x$ — 1

$e^{\int pdx} = e^{2\log(\sec x)} = sec^2 x$ — 1

$ye^{\int pdx} = \int Qe^{\int pdx}\ dx + c$ — 1

$y\ sec^2 x = sec\ x + C$ — 1

$y = 0, \quad x = \frac{\pi}{3}$

$ysec^2 x = sec\ x - 2$ — 1

(OR)

42. b

Total Demand = Total Supply — 1

	D	E	F	G	Supply
A	200 11	50 13	17	14	250
B	16	175 18	125 14	10	300
C	21	24	150 13	250 10	400
Demand	200	225	275	250	

3

Transporation schedule
A→D, A →E, B→E, B→F, C→F, C→G

The Total transportation cost
$= (200x\ 11) + (50\ x\ 13) + (175x\ 18) + (125\ x\ 14) + (150x\ 13) + (250\ x\ 10)$

=₹ 12,200 — 1

43. a

Transition probability matrix T= $\begin{array}{c c} & \begin{array}{cc} A & B \end{array} \\ \begin{array}{c} A \\ B \end{array} & \begin{pmatrix} 0.65 & 0.35 \\ 0.45 & 0.55 \end{pmatrix} \end{array}$ — 1

After one year

(0.15 0.85) $\begin{pmatrix} 0.65 & 0.35 \\ 0.45 & 0.55 \end{pmatrix}$ = (0.48 0.52) — 1

A=48 % , B=52%

At equilibrium — 1

(A B) T=(A B)

(A B) $\begin{pmatrix} 0.65 & 0.35 \\ 0.45 & 0.55 \end{pmatrix}$ = (A B) — 1

0.65A + 0.45 B =A

0.65A+0.45(1-A)=A — 1

A=56 .25% , B=43.75%

(OR)

43. b

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	1				
		7			
2	8		12		
		19		6	
3	27		18		0
		37		6	
4	64		24		0
		61		6	
5	125		30		0
		91		6	
6	216		36		0
		127		6	
7	343		42		
		169			
8	512				

2

$$y = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \ldots\ldots\ldots\ldots$$ — **1**

$When\ \ x = 7.5$

$x_n = 8, \quad h = 1, \quad n = -0.5$ — **1**

$$y = 421.88$$ — **1**

(* Any suitable method can be awarded full marks)

44. a

$\frac{dy}{dx} = \frac{x^2+y^2}{xy}$

sub. y = vx and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ — 1

$$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{vx^2} = \frac{1+v^2}{v}$$ — 1

$\int v dv = \int \frac{dx}{x}$ — 1

$\frac{v^2}{2} = \log x + c$ — 1

$\frac{y^2}{2x^2}$ **= log x +c** — 1

(OR)

44. b

$p = \frac{1}{2} \quad q = \frac{1}{2} \quad n = 4$ — **1**

$P(X = x) = nC_x(p)^x(q)^{n-x}$ — **1**

$P\ (x \geq 1) = \frac{15}{16}$ =0.9375

Number of families with Atleast One boy $= 0.9375 \times 750$
$\cong 703$ — **1**

$P\ (x \geq 2) = \frac{11}{16} = 0.6875$

$Number\ of\ families\ with\ atmost\ 2\ Girls = 0.6875 \times 750\ \cong 516$ — **1**

$P(X = 1) + P(X = 2) + P(X = 3) = 0.875$

$Number\ of\ families\ having$
$Children\ of\ both\ sexes = 0.8750 \times 750 \quad \cong 656$ — **1**

45. a

Month	Year			Monthly Total	Monthly Averages	Seasonal Index
	2001	2002	2003			
January	15	20	18	53	17.67	59.62
February	41	21	16	78	26	87.72
March	25	27	20	72	24	80.97
April	31	19	28	78	26	87.72
May	29	17	24	70	23.33	78.71
June	47	25	25	97	32.33	109.08
July	41	29	30	100	33.33	112.45
August	19	31	34	84	28	94.47
September	35	35	30	100	33.33	112.45
October	38	39	38	115	38.33	129.32
November	40	30	37	107	35.67	120.34
December	30	44	39	113	37.67	127.09

1+1+2

$$\text{S.I.} = \frac{Monthly\ Average}{Grand\ Average} \times 100$$

$$\text{Grand Average} = \frac{355.66}{12} = 29.64$$ — **1**

	(OR)	
45. b	$Mean = E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$ $Mean = E(X) = \frac{1}{2}$ $E(X^2) = \frac{1}{2}$ $V(X) = E(X^2) - (E(X))^2$ $Var(X) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$	**1** **1** **1** **1** **1**
46. a	**R=** $\frac{(-a)}{x+b} - cx + k$ **when x= 0, R = 0** **K=** $\frac{a}{b}$ **R=** $\frac{(-a)}{x+b} - cx + \frac{a}{b}$ **Demand function** **p=**$\frac{R}{x}$ **p=**$\frac{a}{b\,(x+b)} - c$ **x=**$\frac{a}{b\,(p+c)} - b$	1 1 1 1 1
	(OR)	
46. b	$n = 100$ $Sample\ Mean = \bar{x} = 390$ $Population\ mean = \mu = 400$ $Population\ S.D = \sigma = 20$ $Null\ Hypothesis$ $H_0 : \mu =$ **400** $Alternative\ Hypothesis$: $H_1 : \mu \neq$ **400** Z = $(\bar{x} - \mu)/(\sigma/\sqrt{n}) \sim$ N (0,1) Z = -5 |Z| = 5 > Z_α = 2.58 Inference : Null Hypothesis is rejected	**1** **1** **1** **1** **1**

47. a

$$\frac{5x+12}{x^2+5x+6} = \frac{A}{x+3} + \frac{B}{x+2}$$ 1

A=3 , B=2 1

$$\frac{5x+12}{x^2+5x+6} = \frac{3}{x+3} + \frac{2}{x+2}$$

$$\Delta\left[\frac{5x+12}{x^2+5x+6}\right] = \Delta\left[\frac{3}{x+3} + \frac{2}{x+2}\right]$$

$$= \left[\frac{3}{x+4} - \frac{3}{x+3}\right] + \left[\frac{2}{x+3} - \frac{2}{x+2}\right]$$ 2

$$= \frac{-5x-14}{(x+2)(x+3)(x+4)}$$ 1

* Any other alternate method can be awarded full marks

(OR)

47. b

Commodities	Price 2007 (p_0)	Price 2011 (p_1)	Weights (V)	P= $\frac{p_1}{p_0}x100$	PV
A	350	400	40	114.286	4571.44
B	175	250	35	142.857	4999.995
C	100	115	15	115	1725
D	75	105	20	140	2800
E	60	80	25	133.333	3333.325
Total			135		17429.76

3

Cost of Living Index Number = $\frac{\sum PV}{\sum V} =$ **129.1093** 2*